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Task sheets on List

1. Write a Python program to find the maximum and minimum values in a given list.
2. Given a list of integers, find the second-largest element without sorting the list.
3. Write a program to remove duplicate elements from a list while preserving the original order.
4. Given two lists, find the common elements between them and store them in a new list.
5. Write a Python function to check if a list is sorted in ascending order.
6. Given a list of strings, write a program to find the longest string in the list.
7. Implement a function to merge two sorted lists into a single sorted list.
8. Given a list of numbers, write a program to find the sum of all even numbers in the list.
9. Write a Python program to remove all occurrences of a specific element from a list.
10. Given a list of strings, write a program to count the occurrences of each word in the list and store them in a dictionary.

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11. Implement a function to rotate a list by a specified number of positions to the left.
12. Write a program to find the kth smallest element in a given list.
13. Given a list of numbers, write a Python function to find the product of all elements in the list.
14. Write a program to find the index of the first occurrence of a specific element in a list.
15. Given a list of lists, write a Python function to flatten the list into a single list.

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Solutions

#1. Find the maximum and minimum values in a given list:

```
# lst = [1, 5, 3, 8, 2]
# maximum = max(lst)
# minimum = min(lst)
# print("Maximum:", maximum)
# print("Minimum:", minimum)
```

#2. Find the second-largest element in a list without sorting:

```
# lst = [1, 5, 3, 8, 2]
# l = max(lst)
# lst.remove(l)
# l2 = max(lst)
# print(l2)
```

#3. Remove duplicate elements from a list while preserving the original order:

```
# lst = [1, 2, 2, 3, 4, 4, 5]
# unique_list = []
# for item in lst:
#     if item not in unique_list:
#         unique_list.append(item)
#
# print("Unique List:", unique_list)
```

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#4. Find common elements between two lists and store them in a new list:

```
# list1 = [1, 2, 3, 4, 5]
# list2 = [3, 4, 5, 6, 7]
# common_elements = []
#
# for item in list1:
#     if item in list2 and item not in common_elements:
#         common_elements.append(item)
#
# print("Common Elements:", common_elements)
```

#5. Check if a list is sorted in ascending order:

```
# lst = [1, 2, 3, 4, 5]
# lst2= lst.copy()
# if lst == sorted(lst2):
#     print("List 1 is sorted ")
# else:
#     print("List is not sorted")
```

#6. Find the longest string in a list of strings:

```
# lst = ["apple", "banana", "orange", "grapes"]
# longest_string = max(lst, key=len)
# print("Longest String:", longest_string)
```

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#7. Merge two sorted lists into a single sorted list:

```
# list1 = [1, 3, 5, 7, 9]
# list2 = [2, 4, 6, 8, 10]
# list1 += list2
# list1.sort()
# print(list1)
```

#8. Find the sum of all even numbers in a list:

```
# lst = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
# even_sum = sum(num for num in lst if num % 2 == 0)
# print("Sum of Even Numbers:", even_sum)
```

#9. Remove all occurrences of a specific element from a list:

```
# lst = [1, 2, 3, 2, 4, 5, 2]
# element_to_remove = 2
# filtered_list = [num for num in lst if num !=
# element_to_remove]
# print("List after removing", element_to_remove, ":",
# filtered_list)
```

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#10. Count the occurrences of each word in a list and store them in a dictionary:

```
# lst = ["apple", "banana", "orange", "apple",  
"grapes", "banana"]  
# for word in lst:  
#     word_count[word] = word_count.get(word, 0) + 1  
# print("Word Count Dictionary:", word_count)
```

#11. Rotate a list by a specified number of positions to the left:

```
# lst = [1, 2, 3, 4, 5]  
# positions_to_rotate = 2  
# rotated_list = lst[positions_to_rotate:] +  
lst[:positions_to_rotate]  
# print("Rotated List:", rotated_list)
```

#12. Find the kth smallest element in a given list:

```
# lst = [8, 3, 6, 2, 7, 4]  
# k = 3  
# sorted_lst = sorted(lst)  
# kth_smallest = sorted_lst[k - 1]  
# print(f"{k}th Smallest Element:", kth_smallest)
```

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#13. Find the product of all elements in a list:

```
# lst = [1, 2, 3, 4, 5]
# product = 1
# for num in lst:
#     product *= num
# print("Product:", product)
```

14. Find the index of the first occurrence of a specific element in a list:

```
# lst = [1, 2, 3, 2, 4, 5, 2]
# element_to_find = 2
# index = lst.index(element_to_find)
# print("Index of", element_to_find, ":", index)
```

15. Flatten a list of lists into a single list:

```
# list_of_lists = [[1, 2, 3], [4, 5], [6, 7, 8, 9]]
# flattened_list = [num for sublist in list_of_lists
# for num in sublist]
# print("Flattened List:", flattened_list)
```